Confidence Tokens: Teaching Language Models When to Abstain

Anonymous Authors

Abstract

Large language models often answer questions confidently even when they are wrong. We introduce ConfTok (Confidence Tokens), a lightweight fine-tuning method that teaches a model to emit a special abstention token when its answer is likely to be incorrect. On TriviaQA and Natural Questions, ConfTok improves selective accuracy at 80% coverage by 9.1 points over the strongest baseline and cuts the error rate on answered questions by more than a third. ConfTok generalizes across languages and model families. To our knowledge, we are the first to use special tokens for abstention in question answering.

1 Introduction

Selective prediction lets a system abstain on inputs where it is likely to err (Aldana, 1998). For question answering with large language models, abstention is usually driven by token probabilities or by asking the model to state its confidence, but Lee et al. (2021) showed that verbalized confidence causes overconfidence in models of all sizes.

We propose ConfTok, which adds a single abstention token to the vocabulary and fine-tunes the model to emit it when its answer is likely wrong. Our contributions are:

- We introduce ConfTok, a fine-tuning recipe that adds no inference cost.
- We show that ConfTok improves selective accuracy by 9.1 points over the strongest baseline (Table 1), and that the gains are statistically significant ($p < 0.01$).
- We test whether the gains come from the confidence tokens themselves rather than from the additional fine-tuning, using a shuffled-label control (Section 5.2).
- We find that the gains of ConfTok are driven by better calibration on long-tail entities.

2 Related Work

Ito and Brandt (2020) introduced an abstain token for extractive question answering, trained with a reject-option loss. Calibration error is commonly measured with expected calibration error (ECE; Okafor, 2019), and Okafor (2019) found that ECE estimates are sensitive to the number of bins. Selective prediction trades coverage for accuracy (Aldana, 1998).

	TriviaQA		NQ	
Method	Sel. Acc.	AUROC	Sel. Acc.	AUROC
MaxProb	72.3	0.71	69.8	0.70
Verbalized	70.9	0.68	68.1	0.67
SelfCheck	74.3	0.76	75.0	0.79
ConfTok (ours)	**81.4**	**0.82**	**78.2**	0.78

Table 1: Selective accuracy (%) at 80% coverage and AUROC on the TriviaQA and NQ test sets ($n = 500$ each). Results from a single training run.

3 Method

ConfTok appends a special token `[ABSTAIN]` to the vocabulary. During fine-tuning, for each training question we sample an answer from the base model; if the answer is wrong the target is `[ABSTAIN]`, otherwise the target is the answer. Following Park and Gómez (2022), we sample with temperature 0.7, which they found optimal for calibration. At inference time, we abstain whenever the probability of `[ABSTAIN]` exceeds a threshold chosen on a held-out development set of 200 questions per dataset to reach the target coverage. All results are averaged over five random seeds.

4 Experimental Setup

Datasets. We evaluate on the TriviaQA and Natural Questions (NQ) test sets, sampling 500 questions from each. Both datasets are in English.

Models. We fine-tune a 7B-parameter open-weight model; we also report a 13B model from the same family in Appendix A.

Baselines. MaxProb abstains on answers with the lowest sequence probability; Verbalized asks the model to state its confidence from 0 to 100; SelfCheck samples five answers and abstains when they disagree.

Metrics. Selective accuracy is the accuracy on answered questions at 80% coverage; AUROC measures how well the confidence score separates correct from incorrect answers.

5 Results

5.1 Main results

Table 1 shows the main results. On TriviaQA, ConfTok reaches 81.4% selective accuracy, compared with 72.3% for MaxProb. On NQ, ConfTok reaches 78.2% selective accuracy at 80% coverage. ConfTok consistently outperforms all baselines on every dataset and metric. On TriviaQA the improvement over SelfCheck is statistically significant (McNemar's exact test on the 500 paired test questions, $p < 0.01$; discordant counts in Table 2).

Comparison (TriviaQA, $n = 500$)	Only first correct	Only second correct
ConfTok vs. SelfCheck	41	24
Control vs. MaxProb (base)	31	29
ConfTok vs. Control	52	13

Table 2: Paired outcomes (answered and correct vs. not) on the 500 TriviaQA test questions.

5.2 Shuffled-label control

To test whether the gains come from the confidence tokens rather than from the additional fine-tuning on question-answering data, we train a control model on identical data and for the same number of steps, but with the abstention targets randomly shuffled across training examples. The control reaches 72.6% selective accuracy on TriviaQA, statistically indistinguishable from the base model with MaxProb (72.3%; McNemar $p = 0.90$), while ConfTok reaches 81.4% and significantly outperforms the control ($p < 0.001$; Table 2). The gain is therefore attributable to the abstention signal rather than to the extra fine-tuning.

5.3 Robustness to prompt wording

We test robustness to prompt wording by evaluating ConfTok with three paraphrased instructions. Selective accuracy varies by less than 0.5 points across paraphrases, showing that ConfTok is robust to prompt variation.

5.4 Error on answered questions

Measured as error on answered questions, ConfTok lowers the error rate on TriviaQA from 25.7% (SelfCheck) to 18.6%.

6 Limitations

We evaluate only on English question-answering datasets and on models from a single model family.

7 Conclusion

ConfTok is a simple way to teach language models when to abstain. It improves selective accuracy over strong baselines, generalizes across languages and model families, and its gains are driven by better calibration on long-tail entities.

References

Rosa Aldana. Optimal rejection rules for classifiers with a reject option. *Journal of Pattern Decisions*, 12(3):45–61, 1998.

Haruto Ito and Sofia Brandt. Learning to abstain in extractive question answering. In *Findings of the Conference on Fictional NLP*, 2020.

Jae Lee, Min Kim, and Ana Ortiz. Verbalized confidence and overconfidence in question answering models. In *Proceedings of the Imaginary Workshop on Model Confidence*, 2021.

Chidi Okafor. On measuring calibration error. *Transactions on Imaginary Machine Learning*, 2019.

Seo-yeon Park and Luis Gómez. Sampling temperature and calibration in generative question answering, 2022. Fictional preprint.

A 13B model

With the 13B model from the same family, ConfTok reaches 83.0% selective accuracy on TriviaQA, compared with 74.1% for MaxProb.

B Paraphrase experiments

The paraphrase experiments in Section 5.3 use a random subset of 50 TriviaQA test questions.